full_robustness_sensitivity_results

Family	Gate	Scenario	Specification	N	Term	Metric	Estimate	CI_Low	CI_High	p_value	Significant_0_05	Direction	Formatted
Repository-level dependence	Gate 0	Repository-clustered SE	C/S/V + PR size	318	Context	OR	2.14309	1.0902	4.21285	0.0270784	True	positive	2.14* [1.09, 4.21]
Repository-level dependence	Gate 0	Repository-clustered SE	C/S/V + PR size	318	Specificity	OR	65.8384	9.23054	469.604	2.95198e-05	True	positive	65.84*** [9.23, 469
Repository-level dependence	Gate 0	Repository-clustered SE	C/S/V + PR size	318	Verification	OR	0.903855	0.418553	1.95185	0.796908	False	negative	0.90 [0.42, 1.95]
Repository-level dependence	Gate 0	Repository-clustered SE	C/S/V + PR size	318	Log_PR_Size	OR	1.11543	0.906835	1.37202	0.301054	False	positive	1.12 [0.91, 1.37]
Repository-level dependence	Gate 1	Repository-clustered SE	C/S/V + PR size	141	Context	OR	2.21887	1.00231	4.91201	0.0493378	True	positive	2.22* [1.00, 4.91]
Repository-level dependence	Gate 1	Repository-clustered SE	C/S/V + PR size	141	Specificity	OR	0.792625	0.300109	2.09342	0.639061	False	negative	0.79 [0.30, 2.09]
Repository-level dependence	Gate 1	Repository-clustered SE	C/S/V + PR size	141	Verification	OR	8.45432	3.51858	20.3138	1.81723e-06	True	positive	8.45*** [3.52, 20.31
Repository-level dependence	Gate 1	Repository-clustered SE	C/S/V + PR size	141	Log_PR_Size	OR	1.36131	1.0669	1.73695	0.0131089	True	positive	1.36* [1.07, 1.74]
Repository-level dependence	Gate 2	Repository-clustered SE	C/S/V + PR size	89	Context	AME	0.12891	0.0213987	0.236422	0.01877	True	positive	0.129* [0.021, 0.23
Repository-level dependence	Gate 2	Repository-clustered SE	C/S/V + PR size	89	Specificity	AME	-0.0138781	-0.135158	0.107401	0.822539	False	negative	-0.014 [-0.135, 0.10
Repository-level dependence	Gate 2	Repository-clustered SE	C/S/V + PR size	89	Verification	AME	-0.0084639	-0.109619	0.0926906	0.869734	False	negative	-0.008 [-0.110, 0.09
Repository-level dependence	Gate 2	Repository-clustered SE	C/S/V + PR size	89	Log_PR_Size	AME	-0.0371757	-0.077963	0.00361157	0.0740319	False	negative	-0.037 [-0.078, 0.00
Dominant repository sensitivity	Gate 0	Exclude largest repository	C/S/V + PR size	308	Context	OR	1.98122	1.03873	3.77887	0.0379583	True	positive	1.98* [1.04, 3.78]
Dominant repository sensitivity	Gate 0	Exclude largest repository	C/S/V + PR size	308	Specificity	OR	63.0842	8.52229	466.966	4.95196e-05	True	positive	63.08*** [8.52, 466
Dominant repository sensitivity	Gate 0	Exclude largest repository	C/S/V + PR size	308	Verification	OR	0.889567	0.423009	1.87071	0.757665	False	negative	0.89 [0.42, 1.87]
Dominant repository sensitivity	Gate 0	Exclude largest repository	C/S/V + PR size	308	Log_PR_Size	OR	1.09937	0.899223	1.34406	0.355505	False	positive	1.10 [0.90, 1.34]
Dominant repository sensitivity	Gate 0	Exclude top 5 repositories	C/S/V + PR size	188	Context	OR	1.91956	0.984598	3.74234	0.0555697	False	positive	1.92 [0.98, 3.74]
Dominant repository sensitivity	Gate 0	Exclude top 5 repositories	C/S/V + PR size	188	Specificity	OR	53.2761	7.14593	397.198	0.000105077	True	positive	53.28*** [7.15, 397
Dominant repository sensitivity	Gate 0	Exclude top 5 repositories	C/S/V + PR size	188	Verification	OR	0.907773	0.415885	1.98144	0.808039	False	negative	0.91 [0.42, 1.98]
Dominant repository sensitivity	Gate 0	Exclude top 5 repositories	C/S/V + PR size	188	Log_PR_Size	OR	1.09962	0.890221	1.35828	0.378272	False	positive	1.10 [0.89, 1.36]
Dominant repository sensitivity	Gate 1	Exclude largest repository	C/S/V + PR size	135	Context	OR	2.18963	0.945054	5.07325	0.0675284	False	positive	2.19 [0.95, 5.07]
Dominant repository sensitivity	Gate 1	Exclude largest repository	C/S/V + PR size	135	Specificity	OR	0.789568	0.289809	2.15113	0.644057	False	negative	0.79 [0.29, 2.15]
Dominant repository sensitivity	Gate 1	Exclude largest repository	C/S/V + PR size	135	Verification	OR	8.58932	3.55443	20.7562	1.77856e-06	True	positive	8.59*** [3.55, 20.76
Dominant repository sensitivity	Gate 1	Exclude largest repository	C/S/V + PR size	135	Log_PR_Size	OR	1.32064	1.04756	1.6649	0.0186177	True	positive	1.32* [1.05, 1.66]
Dominant repository sensitivity	Gate 1	Exclude top 5 repositories	C/S/V + PR size	121	Context	OR	2.36434	0.972827	5.74623	0.0575433	False	positive	2.36 [0.97, 5.75]
Dominant repository sensitivity	Gate 1	Exclude top 5 repositories	C/S/V + PR size	121	Specificity	OR	0.755377	0.264233	2.15943	0.600646	False	negative	0.76 [0.26, 2.16]
Dominant repository sensitivity	Gate 1	Exclude top 5 repositories	C/S/V + PR size	121	Verification	OR	7.69126	3.03216	19.5094	1.74135e-05	True	positive	7.69*** [3.03, 19.51
Dominant repository sensitivity	Gate 1	Exclude top 5 repositories	C/S/V + PR size	121	Log_PR_Size	OR	1.36428	1.06831	1.74225	0.0127891	True	positive	1.36* [1.07, 1.74]
Dominant repository sensitivity	Gate 2	Exclude largest repository	C/S/V + PR size	83	Context	AME	0.13374	-0.103734	0.371214	0.269678	False	positive	0.134 [-0.104, 0.37
Dominant repository sensitivity	Gate 2	Exclude largest repository	C/S/V + PR size	83	Specificity	AME	-0.0017561	-0.242457	0.238945	0.988591	False	negative	-0.002 [-0.242, 0.23
Dominant repository sensitivity	Gate 2	Exclude largest repository	C/S/V + PR size	83	Verification	AME	-0.016266	-0.236384	0.203852	0.884841	False	negative	-0.016 [-0.236, 0.20
Dominant repository sensitivity	Gate 2	Exclude largest repository	C/S/V + PR size	83	Log_PR_Size	AME	-0.0348446	-0.098538	0.0288495	0.283621	False	negative	-0.035 [-0.099, 0.02
Dominant repository sensitivity	Gate 2	Exclude top 5 repositories	C/S/V + PR size	71	Context	AME	0.0980935	-0.153001	0.349188	0.443862	False	positive	0.098 [-0.153, 0.34
Dominant repository sensitivity	Gate 2	Exclude top 5 repositories	C/S/V + PR size	71	Specificity	AME	0.0528846	-0.20172	0.307489	0.683928	False	positive	0.053 [-0.202, 0.30
Dominant repository sensitivity	Gate 2	Exclude top 5 repositories	C/S/V + PR size	71	Verification	AME	-0.0648588	-0.303635	0.173918	0.59446	False	negative	-0.065 [-0.304, 0.17
Dominant repository sensitivity	Gate 2	Exclude top 5 repositories	C/S/V + PR size	71	Log_PR_Size	AME	-0.039852	-0.108899	0.0291946	0.257953	False	negative	-0.040 [-0.109, 0.02

Family	Gate	Scenario	Specification	N	Term	Metric	Estimate	CI_Low	CI_High	p_value	Significant_0_05	Direction	Formatted
Language sensitivity	Gate 0	Exclude dominant language	C/S/V + PR size	158	Context	OR	1.46592	0.720519	2.98245	0.291219	False	positive	1.47 [0.72, 2.98]
Language sensitivity	Gate 0	Exclude dominant language	C/S/V + PR size	158	Specificity	OR	49.9791	6.64532	375.89	0.000144877	True	positive	49.98*** [6.65, 375
Language sensitivity	Gate 0	Exclude dominant language	C/S/V + PR size	158	Verification	OR	0.732028	0.323457	1.65668	0.454124	False	negative	0.73 [0.32, 1.66]
Language sensitivity	Gate 0	Exclude dominant language	C/S/V + PR size	158	Log_PR_Size	OR	1.0613	0.859679	1.3102	0.579962	False	positive	1.06 [0.86, 1.31]
Language sensitivity	Gate 0	Exclude top 2 languages	C/S/V + PR size	127	Context	OR	1.30253	0.59832	2.83557	0.50547	False	positive	1.30 [0.60, 2.84]
Language sensitivity	Gate 0	Exclude top 2 languages	C/S/V + PR size	127	Specificity	OR	46.3384	6.06066	354.293	0.000218966	True	positive	46.34*** [6.06, 354
Language sensitivity	Gate 0	Exclude top 2 languages	C/S/V + PR size	127	Verification	OR	0.729302	0.292591	1.81783	0.49814	False	negative	0.73 [0.29, 1.82]
Language sensitivity	Gate 0	Exclude top 2 languages	C/S/V + PR size	127	Log_PR_Size	OR	1.16058	0.921293	1.46202	0.206193	False	positive	1.16 [0.92, 1.46]
Language sensitivity	Gate 1	Exclude dominant language	C/S/V + PR size	102	Context	OR	3.2722	1.16991	9.15227	0.0238838	True	positive	3.27* [1.17, 9.15]
Language sensitivity	Gate 1	Exclude dominant language	C/S/V + PR size	102	Specificity	OR	0.524492	0.156367	1.75927	0.295972	False	negative	0.52 [0.16, 1.76]
Language sensitivity	Gate 1	Exclude dominant language	C/S/V + PR size	102	Verification	OR	10.3296	3.64877	29.2431	1.09313e-05	True	positive	10.33*** [3.65, 29.2
Language sensitivity	Gate 1	Exclude dominant language	C/S/V + PR size	102	Log_PR_Size	OR	1.28118	0.985955	1.66479	0.0637219	False	positive	1.28 [0.99, 1.66]
Language sensitivity	Gate 1	Exclude top 2 languages	C/S/V + PR size	79	Context	OR	3.39731	0.997674	11.5686	0.0504369	False	positive	3.40 [1.00, 11.57]
Language sensitivity	Gate 1	Exclude top 2 languages	C/S/V + PR size	79	Specificity	OR	0.580421	0.135241	2.49103	0.464202	False	negative	0.58 [0.14, 2.49]
Language sensitivity	Gate 1	Exclude top 2 languages	C/S/V + PR size	79	Verification	OR	10.87	3.28817	35.9336	9.18473e-05	True	positive	10.87*** [3.29, 35.9
Language sensitivity	Gate 1	Exclude top 2 languages	C/S/V + PR size	79	Log_PR_Size	OR	1.2232	0.899815	1.66281	0.198412	False	positive	1.22 [0.90, 1.66]
Language sensitivity	Gate 2	Exclude dominant language	C/S/V + PR size	59	Context	AME	0.127515	-0.182533	0.437562	0.420194	False	positive	0.128 [-0.183, 0.43
Language sensitivity	Gate 2	Exclude dominant language	C/S/V + PR size	59	Specificity	AME	0.0615353	-0.23889	0.36196	0.688086	False	positive	0.062 [-0.239, 0.36
Language sensitivity	Gate 2	Exclude dominant language	C/S/V + PR size	59	Verification	AME	-0.0579782	-0.321507	0.205551	0.666319	False	negative	-0.058 [-0.322, 0.20
Language sensitivity	Gate 2	Exclude dominant language	C/S/V + PR size	59	Log_PR_Size	AME	-0.0323658	-0.106974	0.0422426	0.395186	False	negative	-0.032 [-0.107, 0.04
Language sensitivity	Gate 2	Exclude top 2 languages	C/S/V + PR size	46	Context	AME	0.0841155	-0.264401	0.432632	0.636182	False	positive	0.084 [-0.264, 0.43
Language sensitivity	Gate 2	Exclude top 2 languages	C/S/V + PR size	46	Specificity	AME	0.0339817	-0.308107	0.37607	0.845632	False	positive	0.034 [-0.308, 0.37
Language sensitivity	Gate 2	Exclude top 2 languages	C/S/V + PR size	46	Verification	AME	-0.0823957	-0.381954	0.217163	0.589816	False	negative	-0.082 [-0.382, 0.21
Language sensitivity	Gate 2	Exclude top 2 languages	C/S/V + PR size	46	Log_PR_Size	AME	-0.0150568	-0.100125	0.070011	0.72866	False	negative	-0.015 [-0.100, 0.07
PR-size outlier sensitivity	Gate 1	Exclude top 1% PR size	C/S/V + PR size	139	Context	OR	2.08292	0.89794	4.83169	0.0874129	False	positive	2.08 [0.90, 4.83]
PR-size outlier sensitivity	Gate 1	Exclude top 1% PR size	C/S/V + PR size	139	Specificity	OR	0.93567	0.331314	2.64244	0.900106	False	negative	0.94 [0.33, 2.64]
PR-size outlier sensitivity	Gate 1	Exclude top 1% PR size	C/S/V + PR size	139	Verification	OR	7.83367	3.22196	19.0463	5.59773e-06	True	positive	7.83*** [3.22, 19.05
PR-size outlier sensitivity	Gate 1	Exclude top 1% PR size	C/S/V + PR size	139	Log_PR_Size	OR	1.40335	1.10236	1.78652	0.00593825	True	positive	1.40** [1.10, 1.79]
PR-size outlier sensitivity	Gate 1	Exclude top 5% PR size	C/S/V + PR size	134	Context	OR	2.25391	0.958594	5.29956	0.0624593	False	positive	2.25 [0.96, 5.30]
PR-size outlier sensitivity	Gate 1	Exclude top 5% PR size	C/S/V + PR size	134	Specificity	OR	0.913981	0.323584	2.58159	0.865185	False	negative	0.91 [0.32, 2.58]
PR-size outlier sensitivity	Gate 1	Exclude top 5% PR size	C/S/V + PR size	134	Verification	OR	7.8077	3.1985	19.059	6.37785e-06	True	positive	7.81*** [3.20, 19.06
PR-size outlier sensitivity	Gate 1	Exclude top 5% PR size	C/S/V + PR size	134	Log_PR_Size	OR	1.38651	1.07654	1.78573	0.0113671	True	positive	1.39* [1.08, 1.79]
PR-size outlier sensitivity	Gate 2	Exclude top 1% PR size	C/S/V + PR size	88	Context	AME	0.133179	-0.101868	0.368226	0.266774	False	positive	0.133 [-0.102, 0.36
PR-size outlier sensitivity	Gate 2	Exclude top 1% PR size	C/S/V + PR size	88	Specificity	AME	-0.0127385	-0.244567	0.21909	0.914237	False	negative	-0.013 [-0.245, 0.21
PR-size outlier sensitivity	Gate 2	Exclude top 1% PR size	C/S/V + PR size	88	Verification	AME	-0.00925981	-0.219391	0.200871	0.931173	False	negative	-0.009 [-0.219, 0.20
PR-size outlier sensitivity	Gate 2	Exclude top 1% PR size	C/S/V + PR size	88	Log_PR_Size	AME	-0.038691	-0.101189	0.0238074	0.224993	False	negative	-0.039 [-0.101, 0.02
PR-size outlier sensitivity	Gate 2	Exclude top 5% PR size	C/S/V + PR size	84	Context	AME	0.147784	-0.0956304	0.391199	0.234065	False	positive	0.148 [-0.096, 0.39
PR-size outlier sensitivity	Gate 2	Exclude top 5% PR size	C/S/V + PR size	84	Specificity	AME	-0.0127463	-0.248322	0.222829	0.915544	False	negative	-0.013 [-0.248, 0.22

Family	Gate	Scenario	Specification	N	Term	Metric	Estimate	CI_Low	CI_High	p_value	Significant_0_05	Direction	Formatted
PR-size outlier sensitivity	Gate 2	Exclude top 5% PR size	C/S/V + PR size	84	Verification	AME	-0.000971922	-0.223174	0.22123	0.99316	False	negative	-0.001 [-0.223, 0.22
PR-size outlier sensitivity	Gate 2	Exclude top 5% PR size	C/S/V + PR size	84	Log_PR_Size	AME	-0.0402186	-0.109481	0.0290438	0.255082	False	negative	-0.040 [-0.109, 0.02
PR-size outlier sensitivity	Axis B integrated	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Context	HR	1.02903	0.805731	1.31421	0.818656	False	positive	1.03 [0.81, 1.31]
PR-size outlier sensitivity	Axis B integrated	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Specificity	HR	1.06444	0.810009	1.39878	0.654102	False	positive	1.06 [0.81, 1.40]
PR-size outlier sensitivity	Axis B integrated	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Verification	HR	0.847656	0.651359	1.10311	0.218766	False	negative	0.85 [0.65, 1.10]
PR-size outlier sensitivity	Axis B integrated	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Log_PR_Size	HR	0.937227	0.875951	1.00279	0.0602108	False	negative	0.94 [0.88, 1.00]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Context	HR	1.62168	0.885729	2.96913	0.117171	False	positive	1.62 [0.89, 2.97]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Specificity	HR	0.633455	0.341034	1.17662	0.148406	False	negative	0.63 [0.34, 1.18]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Verification	HR	1.70702	0.992656	2.93547	0.0531931	False	positive	1.71 [0.99, 2.94]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 1% PR size; cutoff=158144.00; removed=3	Cox PH	218	Log_PR_Size	HR	0.774604	0.669356	0.8964	0.000608321	True	negative	0.77*** [0.67, 0.90]
PR-size outlier sensitivity	Axis B integrated	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Context	HR	1.03575	0.806987	1.32936	0.782657	False	positive	1.04 [0.81, 1.33]
PR-size outlier sensitivity	Axis B integrated	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Specificity	HR	1.08197	0.816667	1.43346	0.583066	False	positive	1.08 [0.82, 1.43]
PR-size outlier sensitivity	Axis B integrated	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Verification	HR	0.815483	0.620227	1.07221	0.144095	False	negative	0.82 [0.62, 1.07]
PR-size outlier sensitivity	Axis B integrated	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Log_PR_Size	HR	0.953612	0.884808	1.02777	0.213794	False	negative	0.95 [0.88, 1.03]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Context	HR	1.56533	0.835372	2.93314	0.161941	False	positive	1.57 [0.84, 2.93]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Specificity	HR	0.632314	0.332598	1.20211	0.161992	False	negative	0.63 [0.33, 1.20]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Verification	HR	1.733	0.996276	3.01453	0.0515598	False	positive	1.73 [1.00, 3.01]
PR-size outlier sensitivity	Axis B Cox hazard	Exclude top 5% PR size; cutoff=40817.00; removed=11	Cox PH	230	Log_PR_Size	HR	0.731339	0.625027	0.855734	9.4638e-05	True	negative	0.73*** [0.63, 0.86]
Gate 2 alternative operationalization	Gate 2	Reuse tertiles	Ordinal logistic	80	Context	Coefficient	0.969038	0.078135	1.85994	0.0330184	True	positive	0.97* [0.08, 1.86]
Gate 2 alternative operationalization	Gate 2	Reuse tertiles	Ordinal logistic	80	Specificity	Coefficient	0.16346	-0.796192	1.12311	0.738497	False	positive	0.16 [-0.80, 1.12]
Gate 2 alternative operationalization	Gate 2	Reuse tertiles	Ordinal logistic	80	Verification	Coefficient	-0.394819	-1.21647	0.426829	0.346293	False	negative	-0.39 [-1.22, 0.43]
Gate 2 alternative operationalization	Gate 2	Reuse tertiles	Ordinal logistic	80	Log_PR_Size	Coefficient	-0.214976	-0.460195	0.0302441	0.0857543	False	negative	-0.21 [-0.46, 0.03]
Gate2 binary high-vs-low	Gate 2	High vs low reuse	Binary logistic	60	Context	OR	2.91747	0.84306	10.0961	0.0909443	False	positive	2.92 [0.84, 10.10]
Gate2 binary high-vs-low	Gate 2	High vs low reuse	Binary logistic	60	Specificity	OR	1.23437	0.345707	4.40739	0.745744	False	positive	1.23 [0.35, 4.41]
Gate2 binary high-vs-low	Gate 2	High vs low reuse	Binary logistic	60	Verification	OR	0.764335	0.218298	2.67619	0.674242	False	negative	0.76 [0.22, 2.68]
Gate2 binary high-vs-low	Gate 2	High vs low reuse	Binary logistic	60	Log_PR_Size	OR	0.760425	0.516874	1.11874	0.164417	False	negative	0.76 [0.52, 1.12]
Aggregate PQS robustness	Gate 0	PQS only	PQS + PR size	218	PQS	OR	2.94439	2.1454	4.04093	2.29569e-11	True	positive	2.94*** [2.15, 4.04]
Aggregate PQS robustness	Gate 0	PQS only	PQS + PR size	218	Log_PR_Size	OR	1.05018	0.877445	1.25691	0.593356	False	positive	1.05 [0.88, 1.26]
Aggregate PQS robustness	Gate 1	PQS only	PQS + PR size	141	PQS	OR	2.70132	1.78583	4.08611	2.52339e-06	True	positive	2.70*** [1.79, 4.09]
Aggregate PQS robustness	Gate 1	PQS only	PQS + PR size	141	Log_PR_Size	OR	1.39301	1.12023	1.73221	0.00287251	True	positive	1.39** [1.12, 1.73]
Aggregate PQS robustness	Gate 2	PQS only	PQS + PR size	89	PQS	AME	0.0171407	-0.077876	0.112158	0.723662	False	positive	0.017 [-0.078, 0.11
Aggregate PQS robustness	Gate 2	PQS only	PQS + PR size	89	Log_PR_Size	AME	-0.0379279	-0.0974745	0.0216186	0.211888	False	negative	-0.038 [-0.097, 0.02
Prompt-length control	Gate 0	Add log(prompt_tokens)	C/S/V + PR size + prompt length	218	Context	OR	2.16152	1.13641	4.11134	0.0187837	True	positive	2.16* [1.14, 4.11]
Prompt-length control	Gate 0	Add log(prompt_tokens)	C/S/V + PR size + prompt length	218	Specificity	OR	60.3944	8.10811	449.857	6.26159e-05	True	positive	60.39*** [8.11, 449
Prompt-length control	Gate 0	Add log(prompt_tokens)	C/S/V + PR size + prompt length	218	Verification	OR	0.834094	0.389925	1.78422	0.640073	False	negative	0.83 [0.39, 1.78]
Prompt-length control	Gate 0	Add log(prompt_tokens)	C/S/V + PR size + prompt length	218	Log_PR_Size	OR	1.11227	0.911245	1.35765	0.295489	False	positive	1.11 [0.91, 1.36]
Prompt-length control	Gate 0	Add log(prompt_tokens)	C/S/V + PR size + prompt length	218	log_prompt_tokens	OR	2.55836	0.30117	21.7325	0.389481	False	positive	2.56 [0.30, 21.73]
Prompt-length control	Gate 1	Add log(prompt_tokens)	C/S/V + PR size + prompt length	141	Context	OR	2.14048	0.909116	5.03968	0.0815297	False	positive	2.14 [0.91, 5.04]

Family	Gate	Scenario	Specification	N	Term	Metric	Estimate	CI_Low	CI_High	p_value	Significant_0.05	Direction	Formatted
Prompt-length control	Gate 1	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	141	Specificity	OR	0.556016	0.194684	1.58798	0.272975	False	negative	0.56 [0.19, 1.59]
Prompt-length control	Gate 1	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	141	Verification	OR	6.9634	2.81381	17.2325	2.69666e-05	True	positive	6.96*** [2.81, 17.23
Prompt-length control	Gate 1	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	141	Log_PR_Size	OR	1.36285	1.07618	1.72587	0.0101901	True	positive	1.36* [1.08, 1.73]
Prompt-length control	Gate 1	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	141	log_prompt_tokens	OR	18.3295	1.82162	184.434	0.0135461	True	positive	18.33* [1.82, 184.4
Prompt-length control	Gate 2	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	89	Context	AME	0.128498	-0.102812	0.359808	0.276241	False	positive	0.128 [-0.103, 0.36
Prompt-length control	Gate 2	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	89	Specificity	AME	-0.00493589	-0.247281	0.237409	0.968158	False	negative	-0.005 [-0.247, 0.23
Prompt-length control	Gate 2	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	89	Verification	AME	-0.00741306	-0.216899	0.202073	0.944705	False	negative	-0.007 [-0.217, 0.20
Prompt-length control	Gate 2	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	89	Log_PR_Size	AME	-0.036678	-0.0978414	0.0244854	0.23986	False	negative	-0.037 [-0.098, 0.02
Prompt-length control	Gate 2	Add log(prompt_tokens)	C/S/V + PR size + prompt_length	89	log_prompt_tokens	AME	-0.0579665	-0.541513	0.42558	0.814243	False	negative	-0.058 [-0.542, 0.42
Hold-out stability	Gate 0	Stratified 80/20 split	Train N=175, Holdout N=43	175	Context	Coefficient direction	0.904959	nan	nan	0.0149389	True	positive	beta=0.905*
Hold-out stability	Gate 0	Stratified 80/20 split	Train N=175, Holdout N=43	175	Specificity	Coefficient direction	4.03769	nan	nan	8.81199e-05	True	positive	beta=4.038***
Hold-out stability	Gate 0	Stratified 80/20 split	Train N=175, Holdout N=43	175	Verification	Coefficient direction	-0.257736	nan	nan	0.54301	False	negative	beta=-0.258
Hold-out stability	Gate 0	Stratified 80/20 split	Train N=175, Holdout N=43	175	Log_PR_Size	Coefficient direction	0.0478501	nan	nan	0.702407	False	positive	beta=0.048
Hold-out stability	Gate 1	Stratified 80/20 split	Train N=113, Holdout N=28	113	Context	Coefficient direction	0.685885	nan	nan	0.136527	False	positive	beta=0.686
Hold-out stability	Gate 1	Stratified 80/20 split	Train N=113, Holdout N=28	113	Specificity	Coefficient direction	-0.0390572	nan	nan	0.945385	False	negative	beta=-0.039
Hold-out stability	Gate 1	Stratified 80/20 split	Train N=113, Holdout N=28	113	Verification	Coefficient direction	2.06749	nan	nan	2.42577e-05	True	positive	beta=2.067***
Hold-out stability	Gate 1	Stratified 80/20 split	Train N=113, Holdout N=28	113	Log_PR_Size	Coefficient direction	0.209699	nan	nan	0.0914981	False	positive	beta=0.210
Hold-out stability	Gate 2	Stratified 80/20 split	Train N=71, Holdout N=18	71	Context	Coefficient direction	0.438578	nan	nan	0.479228	False	positive	beta=0.439
Hold-out stability	Gate 2	Stratified 80/20 split	Train N=71, Holdout N=18	71	Specificity	Coefficient direction	-0.157006	nan	nan	0.800494	False	negative	beta=-0.157
Hold-out stability	Gate 2	Stratified 80/20 split	Train N=71, Holdout N=18	71	Verification	Coefficient direction	0.0543526	nan	nan	0.923554	False	positive	beta=0.054
Hold-out stability	Gate 2	Stratified 80/20 split	Train N=71, Holdout N=18	71	Log_PR_Size	Coefficient direction	-0.188859	nan	nan	0.239433	False	negative	beta=-0.189